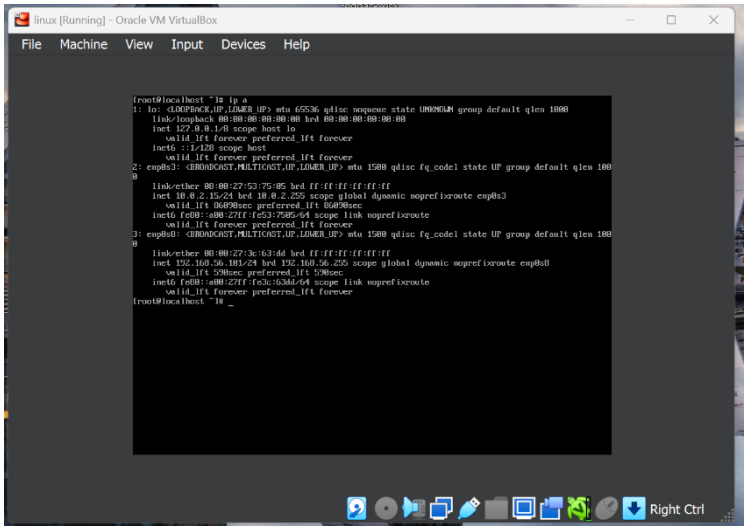
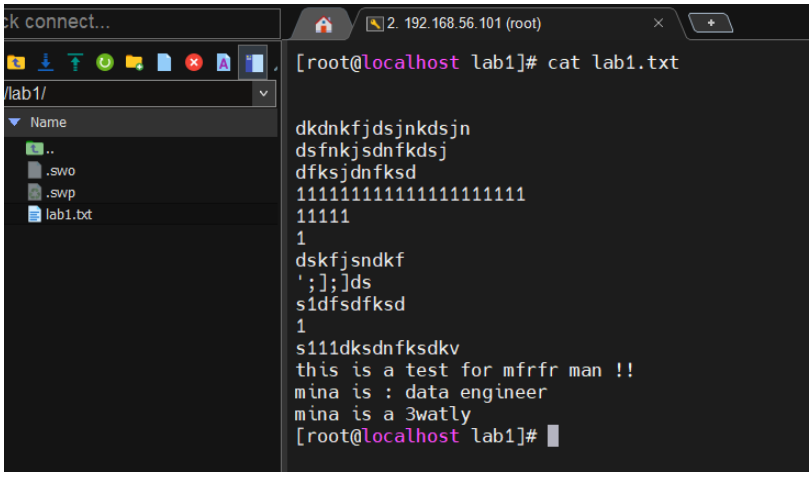
1. Install CentOS /RHEL

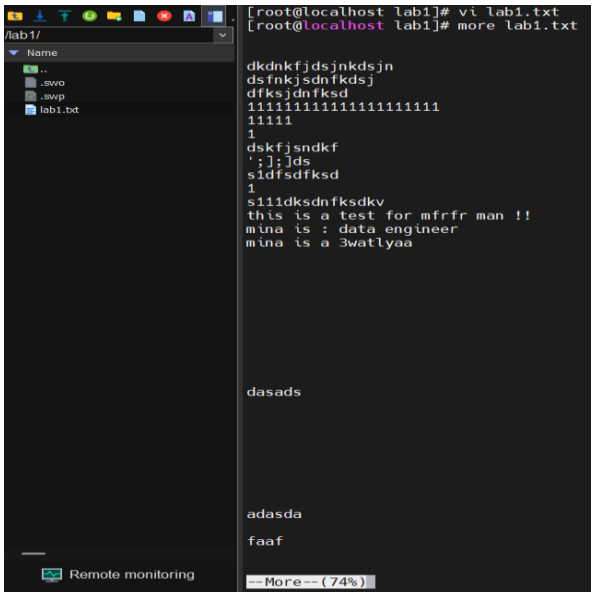


2. What is the difference between cat, more and command?

Cat shows all the file content in screen



More is used to make the file as slides and divide it to more slides like this:



3. What is the difference between rm and rmdir using man?

Rm is responsible for remove files like .txt in my example here

```
[root@localhost lab1]# touch rm_file.txt
[root@localhost lab1]# ls
lab1.txt  rm_file.txt
[root@localhost lab1]# cd rm_file.txt
-bash: cd: rm_file.txt: Not a directory
[root@localhost lab1]# vi rm_file.txt
[root@localhost lab1]# cd rm_file.txt
-bash: cd: rm_file.txt: Not a directory
[root@localhost lab1]# ls
lab1.txt  rm_file.txt
[root@localhost lab1]# rm rm_file.txt
rm: remove regular file 'rm_file.txt'? y
[root@localhost lab1]# ls
lab1.txt
[root@localhost lab1]#
```

Rmdir is used to remove the directory based on the path is right not inside it like that.

Rmdir is responsible for deleting the dirs but when it is empty

```
afs  bin  boot  dev  etc  home  lab1  lib  lib64  media  mnt  opt
[root@localhost /]# cd lab1
[root@localhost lab1]# cd deleteDir
[root@localhost deleteDir]# ls
[root@localhost deleteDir]# rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
[root@localhost deleteDir]# man rmdir
[root@localhost deleteDir]# ls
[root@localhost deleteDir]# rmdir deleteDir
rmdir: failed to remove 'deleteDir': No such file or directory
[root@localhost deleteDir]# cd ..
[root@localhost lab1]# rmdir deleteDir/
[root@localhost lab1]# ls
lab1.txt
[root@localhost lab1]#
```

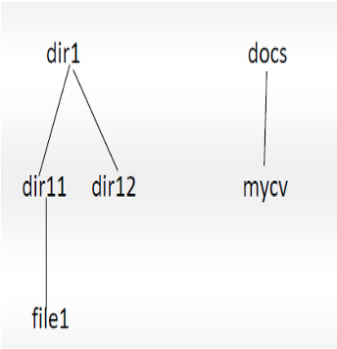
4. Create the following hierarchy under your home directory:

a) Remove dir11 in one-step. What did you notice? And how

did you overcome that?

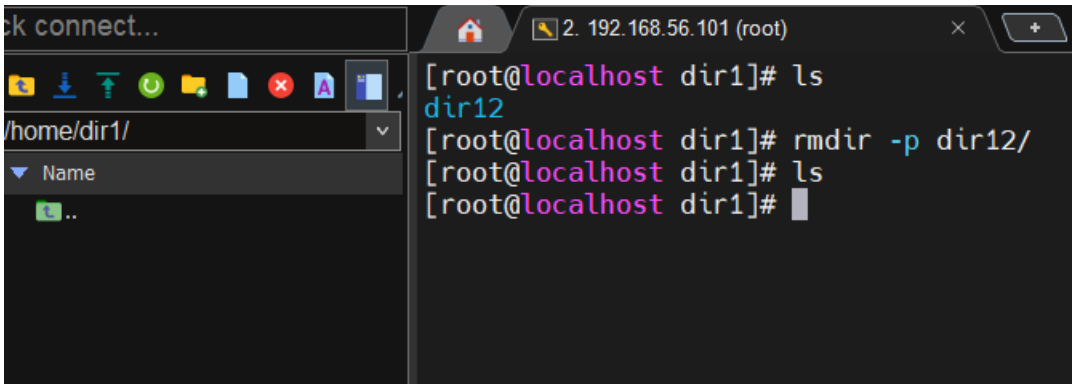
I notice that the dir should removed with rm -r to delete the related files and dir

```
[root@localhost /]# cd home
[root@localhost home]# mkdir dir1 ; cd dir1 ; mkdir dir11 ; mkdir dir12 ; cd dir11 ; touch file1
[root@localhost dir11]# l
-bash: l: command not found
[root@localhost dir11]# ls
file1
[root@localhost dir11]# cd..
-bash: cd..: command not found
[root@localhost dir11]# cd ..
[root@localhost dir1]# cd ..
[root@localhost home]# cd dir1
[root@localhost dir1]# cd ..
[root@localhost home]# cd dir1
[root@localhost dir1]# rm -r dir11
rm: descend into directory 'dir11'? y
rm: remove regular empty file 'dir11/file1'? y
rm: remove directory 'dir11'? y
[root@localhost dir1]#
```

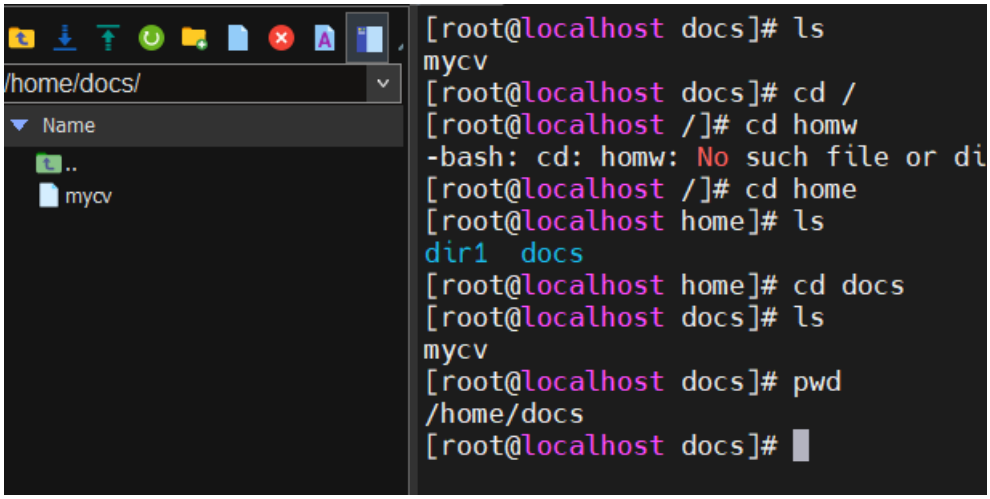


b) Then remove dir12 using rmdir–p command. State what happened to the hierarchy (Note: you are in your home directory).

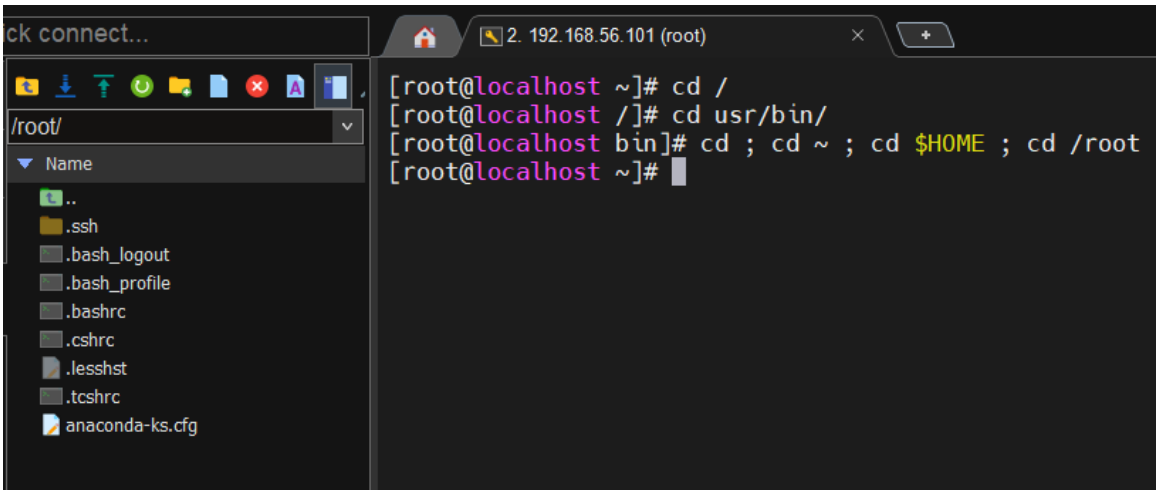
It will delete the dir from the main dir that's because it hasn't no files



c) The output of the command pwd was /home/user. Write the absolute and relative path for the file mycv



7. You are in /usr/bin, list four ways to go to your home directory



8. List Linux commands in /usr/bin that start with letter w

```
[root@localhost ~]# cd usr/bin/
-bash: cd: usr/bin/: No such file or directory
[root@localhost ~]# cd /
[root@localhost /]# cd usr/bin/
[root@localhost bin]# cd /
[root@localhost /]# ls
afs  bin  boot  dev  dir1  etc  home  lab1  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@localhost /]# ls usr/bin/w*
usr/bin/w     usr/bin/wall   usr/bin/watchgnupg  usr/bin/wdctl   usr/bin/whatis.man-db  usr/bin/which     usr/bin/who     usr/bin/write
usr/bin/wait  usr/bin/watch  usr/bin/wc          usr/bin/whatis  usr/bin/whereis        usr/bin/whiptail  usr/bin/whoami
[root@localhost /]#
```

9. Display the first 4 lines of /etc/passwd

```
[root@localhost /]# head -n 4 etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
[root@localhost /]#
```

10. Display the last 7 lines of /etc/passwd

```
[root@localhost /]# tail -n 7 etc/passwd
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
sssd:x:998:995:User for sssd:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/dev/null:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:997:994:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
[root@localhost /]#
```

11 . Display the man page of the passwd file.

```
PASSWD(1)                                User utilities                                PASSWD(1)

NAME
       passwd - update user's authentication tokens

SYNOPSIS
       passwd [-k] [-l] [-u [-f]] [-d] [-e] [-n mindays] [-x maxdays] [-w warndays] [-i inactivedays] [-S] [--stdin] [-?] [--usage] [username]

DESCRIPTION
       The passwd utility is used to update user's authentication token(s).

       This  task is achieved through calls to the Linux-PAM and Libuser API.  Essentially, it initializes itself as a "passwd" service with Linux-PAM and
       utilizes configured password modules to authenticate and then update a user's password.

       A simple entry in the global Linux-PAM configuration file for this service would be:

        #
        # passwd service entry that does strength checking of
        # a proposed password before updating it.
        #
        passwd password requisite pam_cracklib.so retry=3
        passwd password required pam_unix.so use_authtok
        #

       Note, other module types are not required for this application to function correctly.

OPTIONS
       -k, --keep-tokens
              The option -k is used to indicate that the update should only be for expired authentication tokens (passwords);  the  user  wishes  to  keep
              their non-expired tokens as before.

       -l, --lock
              This  option  is  used to lock the password of specified account and it is available to root only. The locking is performed by rendering the
              encrypted password into an invalid string (by prefixing the encrypted string with an !). Note that the account is not  fully  locked  -  the
              user  can  still  log in by other means of authentication such as the ssh public key authentication. Use chage -E 0 user command instead for
              full account locking.

       --stdin
              This option is used to indicate that passwd should read the new password from standard input, which can be a pipe.

       -u, --unlock
              This is the reverse of the -l option - it will unlock the account password by removing the ! prefix. This option is available to root  only.
 Manual page passwd(1) line 1 (press h for help or q to quit)
```